

The Sources and Sinks of Titan: A Literature Review

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1.1 Introduction

Despite the ubiquitous presence of methane on the surface of Titan, the persistence of the compound throughout geologic time has not been yet accounted for (Nixon et al. 2018). By the means of photochemical degradation, the observed supply of liquid methane in lakes should be fully diminished on the order of just 10^5 years, substantially shorter than the age of this Saturnian moon. Over time, methane is irreversibly converted to C_2 and heavier hydrocarbons such as ethane [CH_3]. Continuous photochemical reactions over the lifespan of Titan would both deplete methane reservoirs and cover the surface of Titan in a global ethane ocean. The primary aim of this literature review is to discuss the evidence and implications of primary methane sources and sinks on Titan. Examining these processes will lead to a better understanding of the formation, evolution, and ultimate fate of Titan.

Initially, this paper was to have a research section in which I modeled the subsurface methanifers on Titan. However, this plan was abandoned in favour of literature review due to a lack of information in modeling. Nearly all parameters pertaining to the methane-ethane interactions within the subsurface, surface, and atmosphere of Titan are still very poorly constrained. Any model developed would have likely lacked sufficient evidence for any substantial conclusion. Certain facets of the model would have been arbitrarily constrained, while others omitted entirely. Due to this, I determined that any model I would have pursued would have unacceptable error. For these reasons, this paper is simply a literature review, and does not contain additional independent research.

1.2 Background

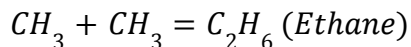
Titan's thick atmosphere (1.5 bar) is predominantly nitrogen and methane, roughly 95 and 5 percent by mole ratio, respectively (Mitri et al. 2006). Due to ultraviolet rays from the sun and high energy particles from Saturn, constituents of Titan's atmosphere are broken down into smaller compounds over time. This process, called photolysis (or photochemical degradation), is the main reason that Titan should be losing methane over time. These broken down molecules may then recombine into a variety of organic chemicals, nitrogen, oxygen, and heavier hydrocarbons. While nitrogen may be easily sustained by this process, methane must be sourced from elsewhere.

Ethane is formed in the upper atmosphere by the recombination of two CH_3 radicals, as described by the chemical processes below (Choukroun & Sotin 2012).

- 1) Photochemical Degradation (Photolysis) of Methane to CH_3 radical and atomic hydrogen



- 2) Recombination of CH_3 radicals into ethane



*Note that it requires two methane compounds to form one ethane molecule

The main takeaway from this set of reactions is that Titan should be bleeding methane over time. However, unless we are witnessing a rare event in which this is the case, there are sources of methane replenishing the degraded particles.

2.1 Methane Sources - Overview

With the constant photochemical decomposition of methane in Titan's atmosphere, the question is often asked as to how this compound is resupplied. For brevity of this review, this section will focus only on cryovolcanism and clathrate hydrates. Clathrates serve as both a source and a sink, so will be referenced in section 3 as well.

2.2 Methane Sources - Cryovolcanism

Cryovolcanism has been hotly contested on Titan for some time, and could easily be the basis for another review entirely. Here, I will outline the evidence concerning cryovolcanism on Titan and discuss its implications for the future of methane on Titan.

The debate for cryovolcanism is centered around the lack of sufficient evidence, as no actual hotspot has been located. However, surface features resembling terrestrial volcanoes have been mapped using instruments onboard Cassini (Lopes et al. 2013). RADAR and Synthetic Aperture Radar (SAR) have been extremely valuable in studying Titan, as the moon's thick haze prohibits the effectiveness of imaging sensors in compiling topographical data. Stereogrammetric RADAR obtained from multiple flybys is used to generate Digital Topographic Models (DTMs), which may be analyzed in the search for cryovolcanoes. DTMs, in addition to SARTopo (topographical data derived from SAR data), have progressed our understanding of Titan's surface features. SARTopo has been used to rule out previous cryovolcano candidates, such as Ganesa Macula. As a circular radar-dark feature with bright edges, Ganesa Macula was originally predicted to be a cryovolcanic dome or shield. However, after more data was collected, SARTopo and DTMs agreed that Ganesa Macula was not an origin of cryovolcanic activity, but

rather “a heavily dissected and eroded region of Titan, with some areas possibly tectonic in origin.”

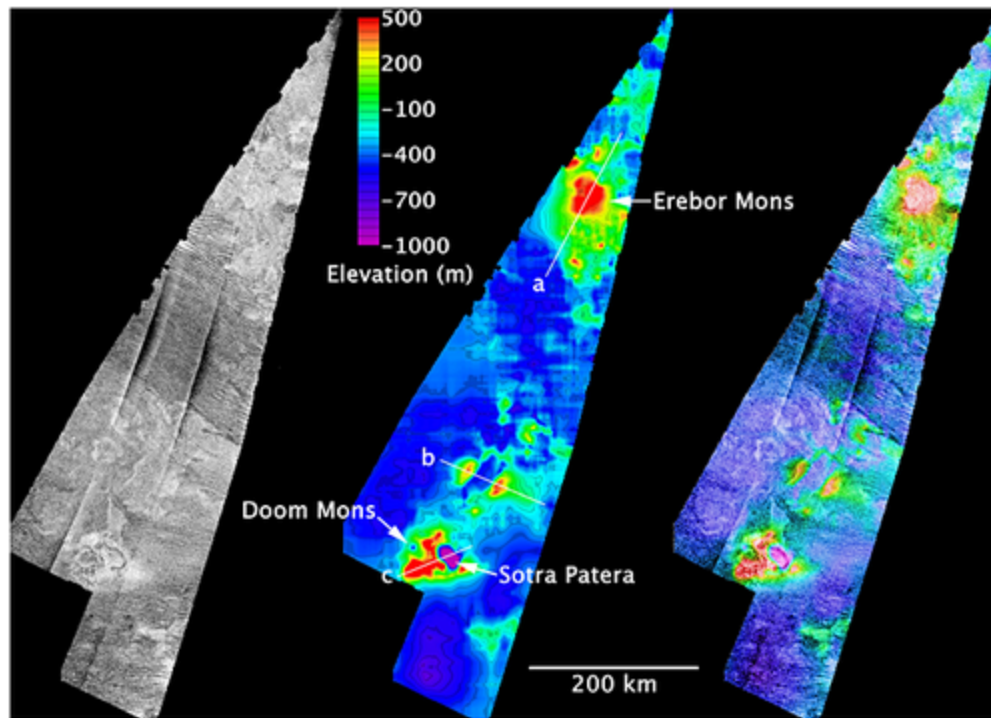


Fig 1.
Cryovolcanic
features on
Titan - Lopes et
al. 2013

However, there are much stronger cases to be made that cryovolcanism does in fact take place on Titan. Using the same SARTopo and DTM data sets, Lopes et al. 2013 studies a region of Sotra Petera that includes surface features Sotra Patera, Mohini Fluctus, Doom Montes, and Erebor Montes. Doom and Erebor Montes are each tall mountains, while Mohini Fluctus is a flow feature emerging from Doom. Sotra Patera is a deep (1.7 ± 0.2 km) depression nearby. The non-circular shape and lack of ejecta blanket means Sotra Patera is likely a caldera or pit rather than an impact. Additionally, the lack of fluvial channels in this region makes a fluvial origin unlikely. Cassini’s Visible and Infrared Mapping Spectrometer (VIMS) data of the region shows compositional differences as compared to nearby plains and dunes. The brightening of the surface in the local region by a factor of two, on a timescale of just one year, shows evidence for

temporal fluctuations in this region. This evidence is indicative of heat flow in the region, and builds a strong argument of the existence of cryovolcanic activity on Titan.

Other than methane clathrates (discussed later in section 3.4), cryovolcanism is the leading mechanism in supplying the atmosphere with methane. Cryovolcanic activity is expected to release the volatile methane from the interior of Titan. While specific processes have yet to be observed, cryovolcanoes are expected to continuously outgas methane on Titan throughout geologic time. Without this interaction, the methane in Titan's atmosphere would depreciate over tens of millions of years. The lack of greenhouse effect would leave the moon in a state referred to as "Snowball Titan" (Wong et al. 2015). This hypothesis allows Titan to normally be in a snowball state, with hydrocarbon rich epochs being triggered by large scale methane outgassing. While we do not know which of these theories is correct, the answer has widespread consequences concerning the past, present, and future of Titan's evolution. However, cryovolcanism is a key factor in either of these scenarios.

2.3 Sources - Clathrate Hydrates

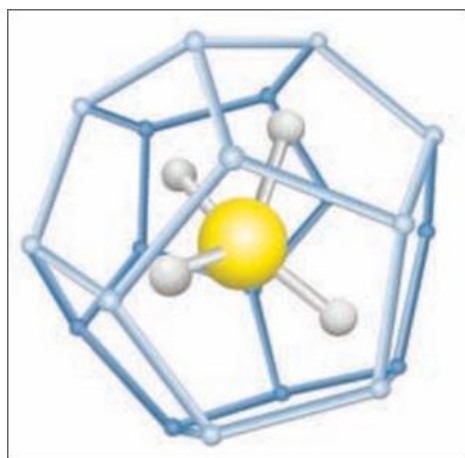


Fig 2. Methane Clathrate - NETL

Clathrate hydrates are water ice compounds that are able to trap another molecule in its open lattice structure (Tobie et al. 2008). Both methane and ethane are compounds that may be trapped to stabilize a water ice lattice. The favourable temperature-pressure conditions of Titan's cold, undifferentiated proto-core were likely to form these

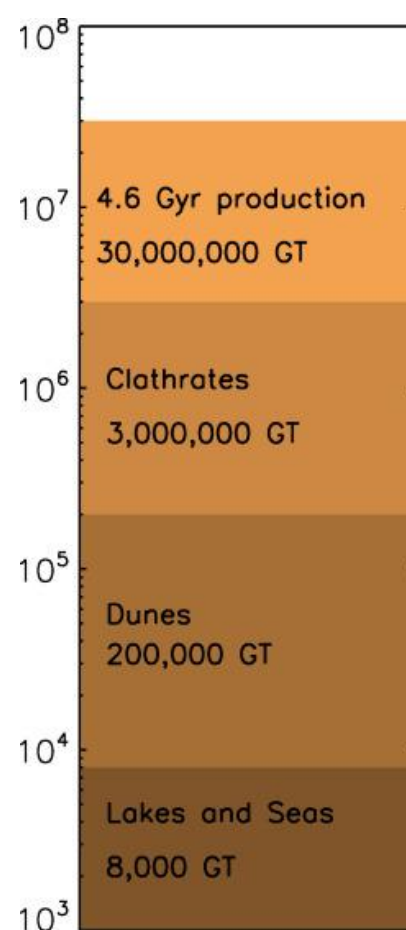
methane clathrates. As photolysis occurs in the upper atmosphere, the resultant ethane descends from the stratosphere, condensing and precipitating onto the surface (Choukroun & Sotin 2012). Ethane is less volatile than methane at the proposed ~10km depth of the clathrate cages, suggesting it replaces methane in this lattice structure. The more volatile methane may then escape to the surface.

Clathrates function to replenish methane to the surface-atmosphere system over time. Each ethane supplants one methane, for a 1:1 ratio of replenishment. If you recall from section 1.2, however, it takes two methane molecules to form one ethane. This is an approximate 50% hydrocarbon recycling. While this process alone cannot sustain methane in the atmosphere, it helps to consistently buffer other systems. This may be the primary supply of methane in between large scale outgassing from cryovolcanoes, which may be few and far between.

3.1 Sinks - Overview

Fig 3. The Sources of Methane. Nixon et al. 2018

In the early 2000's, Cassini's close encounters of Titan helped us understand the moon's surface-atmosphere methane system. As the methane on Titan has been undergoing photochemical degradation for billions of years, it was expected the surface would be covered in a global ethane ocean. While Cassini spotted numerous radar-dark lakes at high latitudes, this volume of hydrocarbons would



only account for about 75,000 years of photochemistry (Nixon et al. 2018). Furthermore, these lakes appear to be mainly methane, and not the expected transmuted ethane sink (Sotin et al. 2012). Equatorial hydrocarbon dunes reminiscent of their sandy analogs in Namibia are an accumulation of up to several hundred million years of solid state ethane. However, this is still orders of magnitudes lower than the expected hydrocarbon sinks, and does not account for any liquid ethane deposits. Subsurface clathrates may serve as both a source of methane and sink for ethane. Choukroun & Sotin in 2012 predicts the seepage of ethane into Titan's crust, which may supplant methane from clathrate cages. This accounts for about 10% of the methane expected to be produced on Titan in its 4.6 billion year life.

3.2 Sinks - Lakes

The most obvious candidate for large methane sinks are of course bodies of water. Liquid methane is thermodynamically allowable everywhere on Titan. That fact coupled with multiple Gya of photolysis led researchers into hypothesizing the existence of global hydrocarbon oceans (Strobel 1982). However, Cassini revealed early this century only relatively small hydrocarbon lakes at polar latitudes (Stofan 2007). Sotin et al. 2012 proposes there is only 4,000-15,000 GT of carbon contained in the lakes. This range is primarily characterized by the ratio of methane to ethane in the lakes, which is still poorly constrained. For reference, this mass of carbon is far less than that contained in the atmosphere, about 200,000 GT. The expected 4.6 Gya of production on Titan is estimated at 30,000,000 GT of carbon. The surface liquid hydrocarbon reserves are far less in volume as compared to Titan's atmospheric reserves. The opposite occurs on Earth, where a much higher volume of water is stored as a liquid rather than in the atmosphere. This may

occur due to the high volatility of methane at most latitudes on Titan. This would also infer that liquid reserves are composed of the more stable ethane, while atmospheric hydrocarbons are primarily methane.

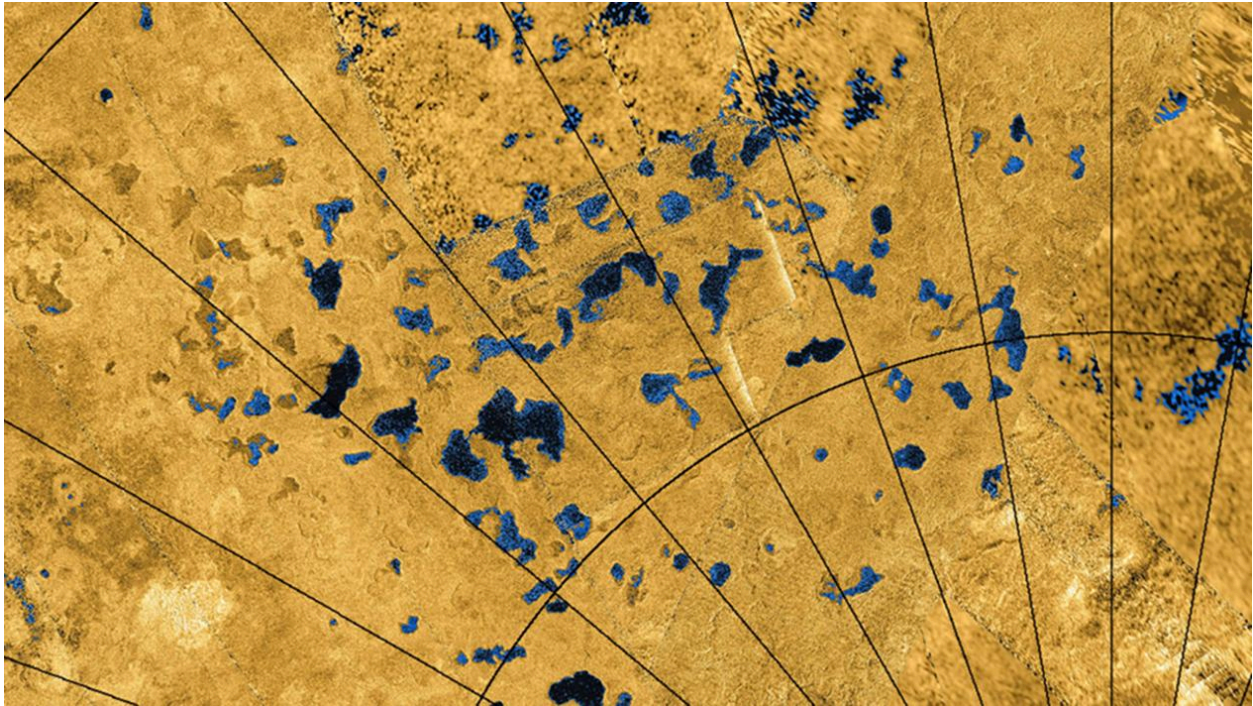


Fig 4. The Lakes of Titan - NASA JPL

3.3 Sinks - Dunes

The equatorial regions of Titan are home to vast dunes, similar in appearance, but not composition, to those in Namibia. The dunes cover approximately 12.5% of Titan's surface, which is about 10 million km², or roughly the size of the United States (Le Gall et al. 2011). Cassini VIMS data suggests the dunes are composed mainly of solid organics. It is proposed the dunes are made from hydrocarbons that have settled from the atmosphere. Dunes are of course in solid form, so may only account for solid hydrocarbon sinks. These dunes account for an

estimated 200,000 GT of carbon. While this is less than a percent of the proposed total carbon production, dunes are still one of the largest sinks we know of to this day.

3.4 Sinks - Methane Clathrates

As previously discussed in section 2.3, methane clathrates serve as a sink for ethane. Each methane is only released when supplanted by an ethane particle. 4.6 Gya of this process is calculated to account for 3,000,000 GT of carbon, or about 10% of the total expected. While this is still an order of magnitude shy of completely accounting for all carbon sinks, clathrates are currently the highest volume of ethane sinks that we understand.

Conclusion

The sources and sinks of methane and ethane on Titan are still yet to be well understood. Further missions must closely examine the lakes, evidence for cryovolcanoes, the composition and formation of dunes, and the interactions of clathrate supplantation on Titan. Our current pool of knowledge on Titan is a commendable start, but only scratches the surface of the complex mechanisms occurring on this foreign moon. Cassini brought upon the golden era of discovery for Titan, but that mission has since ended. Further missions to this moon of Saturn are required to more fully understand any of its processes, and the moon as a whole.

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